

UNIVERSITY OF BUEA

Department of Computer Engineering

CEF 440

Internet and Mobile Programming

TASK 2: REQUIREMENT GATHERING REPORT

**Project: Design and Implementation of a Mobile Application
for Car Fault Diagnosis**

Group 12 Project Report Task 2

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1. Introduction

Requirement gathering is a foundational activity in the software engineering lifecycle. For a project of this nature — an AI-powered mobile application for car fault diagnosis — it is especially critical to ensure that the system being designed aligns closely with real user needs, technical feasibility, and operational constraints.

This report documents the requirement gathering activities performed for Task 2 of the project. It covers the identification of stakeholders, the techniques used to elicit requirements, the nature of data collected and how it was processed, and an assessment of anticipated user reluctance toward the proposed system. Together, these activities form a comprehensive basis for the subsequent design and implementation tasks.

2. Stakeholder Identification

A stakeholder is any individual, group, or organization that has a direct or indirect interest in the system being developed, or whose activities will be affected by it. The identification of stakeholders is essential because different groups have different expectations, needs, and constraints that must all be reflected in the system requirements.

2.1 Primary Stakeholders

Primary stakeholders interact directly with the system.

Stakeholder	Role	Primary Interest
Car Owners / Everyday Drivers	End users of the mobile application	Interpret dashboard warning lights and engine sounds without needing a mechanic
Mobile Application Developers	System builders	Clear, implementable functional and non-functional requirements
UX/UI Designers	Interface architects	Understanding of user demographics, literacy levels, and interaction preferences

2.2 Secondary Stakeholders

Secondary stakeholders do not use the system directly but have significant influence over its development or success.

Stakeholder	Role / Interest
Automobile Manufacturers (OEMs)	May provide standardized OBD-II symbol definitions and technical documentation

Stakeholder	Role / Interest
YouTube Content Creators / Car Repair Experts	Providers of tutorial video content linked in the app's repair guidance module

2.3 Tertiary Stakeholders

These are parties affected indirectly, such as road safety authorities (who benefit from fewer vehicle failures on roads), environmental agencies (fewer undetected emission faults), and the general public who share roads with the vehicles being diagnosed.

3. Requirement Gathering Techniques

Multiple complementary techniques were employed to ensure breadth and depth in the requirements elicited. No single technique is sufficient on its own; a triangulated approach reduces the risk of missing critical requirements.

3.1 Surveys

A structured questionnaire was distributed to a sample of car owners, drawn from university staff, students with vehicles, and community members in the Buea municipality. The survey was administered both online (via Google Forms) and on paper to account for varying levels of digital access.

Key survey objectives:

- Determine the frequency with which drivers encounter unfamiliar warning lights.
- Assess prior exposure to mobile diagnostic apps.
- Gauge willingness to use an AI-based audio diagnostic feature.
- Identify preferred language and notification styles.
- Determine offline usage requirements.

Survey sample characteristics: 80 respondents; aged 18–55; 60% male, 40% female; vehicle age range: 2–20 years; approximately 65% reported encountering a warning light they could not immediately interpret.

3.2 Interviews

Semi-structured interviews were conducted with five automobile mechanics and three car owners with above-average vehicle knowledge. Interviews lasted between 30 and 60 minutes and were recorded (with consent) for later analysis.

Interview themes included:

- Most common diagnostic errors made by non-expert drivers.
- Limitations of current OBD-II scanner tools in non-workshop settings.
- Frequency and variety of abnormal engine sounds encountered in practice.
- Expectations around urgency classification of detected faults.

Notable findings from interviews: Mechanics noted that drivers routinely ignore the check engine light until secondary symptoms appear, often worsening the underlying fault. They also emphasized that the same warning symbol can indicate vastly different fault severities depending on vehicle make and year, which has implications for model training.

3.3 Brainstorming

A facilitated brainstorming session was conducted among the project development team. The session followed a structured format: problem decomposition, divergent idea generation, and convergent prioritisation using dot voting.

Core brainstorming outputs:

- Identification of the core feature modules: Camera Scanner, Audio Recorder, Fault Explainer, Urgency Estimator, and Video Tutorial Linker.
- Recognition that offline-first architecture is necessary given network unreliability in the target use context.
- Agreement on React Native as the candidate framework to maximise cross-platform coverage.
- Proposal to use a hybrid diagnosis strategy: on-device ML models for speed, with cloud API fallback for complex cases.

3.4 Reverse Engineering

Three existing automotive diagnostic applications — Car Scanner ELM OBD2, FIXD, and AutoDoc — were analysed through reverse engineering. This involved installing and operating the applications, examining their feature sets, interaction flows, and error handling, and mapping them against the proposed system.

Reverse engineering findings:

- All three applications require a Bluetooth OBD-II dongle, creating a hardware dependency that the proposed application explicitly avoids.
- None of the reviewed applications support audio-based engine fault detection; this represents a clear differentiator for the proposed system.
- FIXD provides urgency ratings for warning lights — this pattern is worth adopting, but the urgency logic should be trained on local driving conditions.
- AutoDoc's video tutorial integration model (YouTube links per fault code) is a strong reference for the proposed in-app tutorial module.

3.5 Document Analysis

Relevant technical and standards documentation was reviewed, including:

- OBD-II standard (SAE J1979) for standardized diagnostic trouble codes (DTCs) and dashboard symbol definitions.
- ISO 11898 (CAN bus protocol) for understanding how ECU data is structured.
- Published datasets: the ESC (Engine Sound Classification) dataset and the ARCA (Audio Recognition for Car Anomalies) dataset, used to understand training data characteristics.

- Google’s Material Design guidelines and Apple’s HIG documentation, to inform accessibility and UI requirement derivation.

4. Data Gathering

The system requires two primary categories of data for its AI components: dashboard warning light images and engine audio recordings. Data gathering was approached from both existing datasets and primary collection.

4.1 Dashboard Warning Light Image Data

Sources

- OBD-II Symbol Libraries: Public domain SVG and PNG symbol sets conforming to ISO 2575 (symbols for controls, indicators, and tell-tales).
- Web Scraping: Images of real dashboard warning lights photographed under various conditions (partial occlusion, glare, different dashboard colours) were collected from automotive forums and social media with Creative Commons licensing.
- Primary Photography: Team members photographed dashboards of ten different vehicle models (Toyota Corolla, Camry, Hilux; Honda Civic, CRV; Hyundai i10; Kia Picanto; Nissan Almera; Ford Ranger; Mitsubishi Pajero) under varying lighting conditions.

Data Category	Volume Collected	Format	Source Type
OBD-II standard symbols	87 unique symbols	SVG / PNG	ISO standard library
Real dashboard photos	1,240 images	JPEG	Web scraping + primary
Annotated bounding boxes	1,240 entries	JSON (COCO format)	Manual annotation
Engine audio recordings	3,600 clips (10–30s)	WAV (44.1 kHz, mono)	Datasets + primary
Fault-audio label pairs	3,600 entries	CSV	Expert annotation

4.2 Engine Audio Data

Sources

- Public Datasets: The AudioSet (Google DeepMind) vehicle sub-class was filtered for car engine clips. The ESC-50 environmental sound dataset contributed additional mechanical sound samples.
- Primary Recording: A mechanic partner recorded engine sounds for six common fault types across five vehicle models using a calibrated smartphone microphone at 30 cm from the engine bay. Fault types covered: normal idle, engine knocking, belt squealing, exhaust hissing, timing chain rattle, and starter motor failure.

4.3 User Behaviour Data

Survey responses and interview transcripts were collected as structured data. Responses were exported to CSV from Google Forms. Interview recordings were transcribed and coded thematically using open coding in NVivo-compatible CSV exports.

5. Data Cleaning

Raw collected data contained noise, inconsistencies, and quality issues that required systematic resolution before use in model training and requirements specification.

5.1 Image Data Cleaning

- **Duplicate Removal:** Perceptual hashing (pHash) was applied to the image corpus. 143 near-duplicate images were identified and removed, reducing the dataset from 1,240 to 1,097 images.
- **Resolution Filtering:** Images below 224×224 pixels were discarded as insufficient for feature extraction by convolutional models. 67 images were removed on this basis.
- **Annotation Correction:** Bounding box coordinates were validated for consistency. 38 annotations had mismatched class labels (e.g., “engine temperature” labelled as “oil pressure”); these were corrected via a second-pass expert review.
- **Lighting Normalisation:** Images with extreme overexposure (histogram saturation $> 95\%$) or underexposure (mean pixel value < 30) were augmented using CLAHE (Contrast Limited Adaptive Histogram Equalization) rather than discarded, to improve model robustness.
- **Class Imbalance Identification:** The cleaned dataset revealed that 12 of 87 fault classes had fewer than 10 samples. These were flagged for synthetic augmentation using image transformation pipelines.

5.2 Audio Data Cleaning

- **Silence Trimming:** Clips with more than 3 consecutive seconds of silence at the start or end were trimmed using a voice activity detection (VAD) threshold of -40 dBFS.
- **Background Noise Assessment:** Signal-to-noise ratio (SNR) was computed for all clips. Clips with SNR below 10 dB (indicative of excessive ambient noise) were either bandpass-filtered or removed (214 clips removed; 87 filtered).
- **Resampling:** All audio files were resampled to a uniform 22,050 Hz sampling rate and converted to mono for consistency with model input requirements.
- **Label Verification:** Audio clips labelled by dataset providers were cross-checked against descriptions by the mechanic partner. 52 misclassified clips (primarily confusing “belt squeal” with “tyre squeal”) were relabelled.
- **Duration Normalisation:** Clips were padded or trimmed to a uniform 10-second duration using zero-padding and windowed truncation respectively.

5.3 Survey Data Cleaning

- **Incomplete responses** with more than 3 unanswered questions (out of 18) were excluded (9 records removed).
- **Outlier responses** (e.g., reported driving for 50 years at age 22) were flagged and verified; 3 were corrected via follow-up, 2 were removed.
- **Free-text responses** were spell-checked and normalised to standard terminology (e.g., “check engine lamp” $\rightarrow$ “check engine light”).

5.4 Cleaned Dataset Summary

Dataset	Raw Count	Cleaned Count	Reduction (%)
Dashboard images	1,240	1,030	16.9%
Audio clips	3,600	3,299	8.4%
Survey responses	80	69	13.8%
Interview coded segments	N/A	214 codes	—

6. User Reluctance Assessment

User reluctance assessment evaluates the extent to which potential users may resist adopting or fully engaging with the proposed system. Identifying sources of reluctance early allows the team to design mitigations directly into the product.

6.1 Identified Sources of Reluctance

6.1.1 Privacy Concerns

Survey results indicated that 41% of respondents expressed concern about the application accessing the camera and microphone. This is heightened by the sensitivity surrounding vehicle location and usage patterns that the application could potentially infer.

Mitigation: The application will implement explicit permission prompts with plain-language explanations of why each permission is needed and how data is used. All audio and image data processed by on-device models will not be transmitted to external servers by default.

6.1.2 Distrust of AI Accuracy

A significant proportion of interviewees (4 out of 5 mechanics and 2 out of 3 experienced car owners) expressed scepticism that an AI system could reliably diagnose engine faults from audio alone. The primary concern was the high variability of engine sounds across vehicle makes, models, and ages.

Mitigation: The application will present diagnoses as “likely fault candidates” with associated confidence scores rather than definitive determinations. A disclaimer will direct users to consult a professional for high-urgency faults. Model performance benchmarks will be displayed in the onboarding screen to build credibility.

6.1.3 Digital Literacy Barriers

The target demographic includes drivers of varying educational backgrounds and varying degrees of smartphone proficiency. 28% of survey respondents indicated difficulty with “complex app interfaces”. Among respondents over 45, this rose to 54%.

Mitigation: UI design will prioritize large touch targets, voice-guided instructions, minimal text on primary action screens, and a simplified onboarding flow. The application will support both English and French to address Cameroon’s bilingual context.

6.1.4 Cost and Storage Concerns

18% of respondents mentioned concern about the app consuming mobile data or requiring expensive in-app purchases. Additionally, 23% expressed concern about storage requirements for offline datasets.

Mitigation: The application will offer a free tier with core diagnostic functionality. Offline model files will be compressed and modular: users will be prompted to download only models relevant to their vehicle type. The baseline install size will be kept below 80 MB.

6.1.5 Reliability and Safety Trust

Several respondents noted that incorrect diagnosis could lead to drivers continuing to operate unsafe vehicles or, conversely, causing unnecessary alarm. This creates a safety-trust paradox: under-confidence in the app may cause it to be ignored, while over-confidence may cause harm.

Mitigation: Urgency levels will be calibrated conservatively (i.e., erring toward higher urgency when uncertain). All critical-level fault outputs will include a non-dismissible prompt recommending immediate professional inspection.

6.2 Reluctance Summary Matrix

Reluctance Factor	Prevalence	Severity	Planned Mitigation
Privacy / data concerns	41%	High	On-device processing; explicit permission UX
Distrust of AI accuracy	~67% (expert users)	High	Confidence scores; professional referral prompts
Digital literacy barriers	28–54%	Medium	Simplified UI; bilingual support; voice guidance
Cost / data / storage concerns	18–23%	Medium	Free tier; modular offline models; <80 MB base install
Safety over/under-confidence	Qualitative	High	Conservative urgency calibration; mandatory warning for critical faults

7. Consolidated Requirements Summary

Based on the stakeholder analysis, elicitation activities, and reluctance assessment, the following high-level requirement categories have been identified. These will be refined into formal functional and non-functional requirements in Task 3.

7.1 Functional Requirements (High-Level)

- FR-01: The system shall capture and classify dashboard warning light symbols using the device camera.
- FR-02: The system shall detect and classify engine fault types from audio recordings.
- FR-03: The system shall support simultaneous detection of multiple active warning lights.

- FR-04: The system shall provide explanations, probable causes, and urgency ratings for each detected fault.
- FR-05: The system shall link users to relevant YouTube tutorial videos from verified automotive professionals.
- FR-06: The system shall operate core diagnostic features in offline mode using pre-downloaded models.
- FR-07: The system shall synchronise with online fault databases when network connectivity is available.

7.2 Non-Functional Requirements (High-Level)

- NFR-01: The application shall complete visual fault classification within 3 seconds of image capture.
- NFR-02: The application shall achieve a minimum of 85% accuracy on the validation dataset for both image and audio classification tasks.
- NFR-03: The offline base install shall not exceed 80 MB.
- NFR-04: The application shall be deployable on Android 9.0+ and iOS 13.0+ platforms.
- NFR-05: The application shall comply with GDPR principles regarding data minimisation and purpose limitation.

8. Conclusion

This report has presented a comprehensive account of the requirement gathering activities undertaken for the Car Fault Diagnosis Mobile Application project. Through a combination of surveys, interviews, brainstorming, reverse engineering, and document analysis, a rich and multi-faceted set of requirements has been elicited from a broad stakeholder base.

The data gathering phase produced a substantial corpus of labelled images and audio recordings, which have been cleaned and prepared for model development. The user reluctance assessment has surfaced five critical adoption barriers — privacy, accuracy trust, digital literacy, cost, and safety confidence — each of which has been mapped to a concrete design or policy mitigation.

The outputs of this task provide a solid foundation for Task 3 (System Design) and Task 4 (Implementation), ensuring that the resulting system is technically grounded, user-centred, and responsive to the real constraints of its intended deployment context.